INCH-POUND

MIL-T-52864D
6 September 1990
SUPERSEDING
MIL-T-52864C
23 March 1990
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22 April 1987

MILITARY SPECIFTCATION

TRUCKS, LIFT, FORK: DUSEL-ENGINE-DRIVEN, PNEUMATIC-TIRED,

10,000, 15,000 and 20,000 POUND CAPACITY;

GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for diesel-engine-driven pneumatic-tired depot forklift trucks used for handling cargo in and around warehouses, loading platform, and docks.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

GG-P-455	- Plates and Foils, Photographic (Photosensitive anodized Aluminum).
W-F-800	- Fuel Oil, Diesel.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-T-704	- Treatment and Painting of Material.
NIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-G-3859	- Grease Gun, Hand-Operated, Lever, Push and Screw Type.
MIL-E-10062	- Engine, Preparation for Shipment and Storage of.
MIL-C-46168	- Coating, Alphatic, Polyurethane Chemical Agent Resistant.
MIL-B-46176	- Brake, Fluid, Silicone, Automotive, All Weather, Operational and Preservative.
MIL-E-52649	- Engine Cold Starting Aids, Ether Fuel Primers.
MIL-P-53030	- Primer Coating Epoxy Water Reducible Lead and Chromate Free.
MIL-C-53039	- Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant.
MIL-V-81940	- Valve, Sampling and Bleed, Hydraulic, Type II Systems.
MIL-T-83133	- Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.

STANDARDS

MILTTARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-162	- Materials Handling Equipment: Preparation for Shipment, Storage, Cyclic Maintenance, Routine Testing and Processing.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1474	- Noise Limits for Army Materiel.
MS24207	- Coupling, Grease Gun; Hydraulic Type Nozzle.
MS51336	- Lunette-Coupler Drawbar, Ring.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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DRAWINGS

TACOM
12276986 - Kit, M16 Rifle Mounting.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Tank Automotive Command, ATTN: AMSTA-M, Warren, MI 48397-5000.)

DEPARTMENT OF LABOR (DOL)

29CFR1910 - Occupational Safety and Health Administration.

(Application for copies should be addressed to a local DOL officer or the Superintendent of Documents, US Government Printing Office, Washington, DC 20402).

THE MILITARY TRAFFIC MANAGEMENT COMMAND/(MTMC/TEA)

- Pamphlet 70-1-Transportability for Better Strategic Mobility.

(Application for copies should be addressed to: The Military Traffic Management Command Transportation Engineering Agency (MTMC/TFA), ATTN: MTT-TR, P.O. Box 6276, Newport News, Va. 23606-0276.)

TECHNICAL INFORMATION PACKAGE (TIP)

TIP-52864

(Application for copies should be addressed to the Commanding Officer, US Army Belvoir Research, Development and Engineering Center, ATTN: STRBE-FM, Fort Belvoir, Virginia 22060-5606.)

2.2 Other publication. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-Government documents which is current on the date of the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

B 56.1 - Safety Standard for Powered Industrial Trucks.
B 56.11.3 - Load Handling Symbols for Powered Industrial Trucks.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN NATIONAL STANDARDS INSTITUTE INC. (ANSI)

Z 26.1 - Safety Glazing Materials for Glazing Motor Vehicles Operating on Land Highways.

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(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10017.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE J53	- Minimum Performance Criteria for Emergency Steering of Wheeled Earthmoving Construction Machines.
SAE J96	- Flashing Warning Lamp for Industrial Equipment.
SAE J154	- Operator Enclosures Human Factor Design Considerations.
SAE J180	- Electrical Charging Systems for Construction and Industrial Machinery.
SAE J185	- Access Systems for Off-Road Machines.
SAE J198	- Windshield Wiper Systems, Trucks, Buses, and Multipurpose Vehicles.
SAE J209	- Instrument Face Design and Location for Construction and Industrial Equipment.
SAE J297	- Operator Controls on Industrial Equipment.
SAE J534	- Lubrication Fittings.
SAE J537	- Storage Batteries.
SAE J541	- Voltage Drop for Starting Motor Circuits.
SAE J551	- Performance Levels and Methods of Measurements of Electromagnetic Radiation from Vehicles and Devices (20-1000 MHz).
SAE J585	- Tail Lamps (Rear Position Lamps).
SAE J586	- Stop Lamps for Use on Motor Vehicles Less Than 2032 mm in Overall Length.
SAE J598	- Sealed Lighting Units for Construction and Industrial Machinery.
SAE J614	- Engine and Transmission Dipstick Marking.
SAE J759	- Lighting Identification Code.
SAE J833	- USA Human Physical Dimensions.
SAE J845	- 360 Degree Warning Lamp for Authorized Emergency, Maintenance and Service Vehicles.
SAE J898	- Control Locations for Off-Road Work Machines.
SAE J899	- Operator's Seat Dimensions for Off-Road Self-Propelled Work Machines.
SAE J917	- Marine Push-Pull Control Cables.
SAE J925	- Minimum Service Access Dimensions for Off-Road Machines.
SAE J985	- Vision Factors Considerations in Rear View Mirror Design, Information Report.
SAE J1152	- Braking Performance – Rubber-Tried Construction Machines.
SAE J1362	- Symbols for Controls, Indicators, and Tell-Tales for Off Road Self-Propelled Work Machines.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15086-0001.)

TIRE AND RIM ASSOCIATICH, INC. (TRA)

Tire and Rim Association Yearbook

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(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

UNDERWRITERS LABORATORIES INC. (UL)

UL 558 Standard for Industrial Trucks, Internal Combustion Engine-Powered.

(Application for copies should be addressed to Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for specifications sheets), the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The truck shall be a diesel-engine-driven, front wheel drive, rear wheel steer, forklift truck. The truck shall be the manufacturer's current commercial product and shall have been in production, marketed and in use for a minimum of one year preceding the solicitation for procurement. The introduction of product improvement changes in this one year period is acceptable.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The forklift truck shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the forklift truck may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Recovered material. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and

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reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the forklift truck may be newly fabricated from recovered materials to the maximum extent practicable, provided the forklift truck produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the forklift truck.

3.4 Environmental requirements. The engine shall start without preheating within 5 minutes and the truck shall operate as specified herein within 15 minutes after engine start in any ambient temperature from -0 to 120 °F. When specified, see 6.2, ambient temperature required will be from -25 to 120 °F. Glow plugs to aid in starting are acceptable in lieu of priming system specified in 3.7.1.1.

3.5 Safety. The truck shall conform to the requirements of ANSI B 56.1, UL 558, type D and OSHA standards in effect at the time of manufacture. When specified (see 6.2), the truck shall conform to the requirements of UL 558, type DS and OSHA standards in effect at the time of manufacture. For a DS type truck, the ambient temperature will be from 0 to 120 °F. For a DS type truck, the contractor shall submit to the contracting officer or designated representative, satisfactory evidence that the truck furnished under this specification meet or exceed the requirements of UL 558. Acceptable evidence of meeting these requirements shall be certified test reports from recognized independent testing laboratories acceptable to the Government, indicating that the truck conform to the technical requirements of UL 558. Also, the truck shall be furnished with a safety nameplate as specified herein.

3.5.1 Nuclear safety. When nuclear safety is specified (see 6.2), the truck shall be equipped with additional safety features as specified herein.

3.5.2. Fire extinguisher. When nuclear safety is specified (see 6.2), a fire extinguisher shall be mounted on each truck in an easily accessible location. The fire extinguisher shall have a minimum UL rating of 5-B :C.

3.5.3 Rearview mirror. When specified (see 6.2), the truck shall be equipped with a rearview mirror (with mounting arm) mounted on the left-hand side within the operator's field of view as described in SAE J985.

3.6 Human factors engineering. The characteristics of the truck shall provide for operation by the 5th to 95th percentile male and female in accordance with SAE J833. Controls and displays shall be labeled with universal symbols in accordance with SAE J1362. Controls shall be located in accordance with SAE J898 and SAE J297. Instrumentation shall be in accordance with SAE J209.

3.7 Engine. The engine shall be a commercial type water cooled diesel engine. The engine furnished shall operate on fuel conforming to both VV-F-800, grade DF2 and MIL-T-83133. When specified (see 6.2) the first article test shall include a test for operation with MIL-T-83133.

3.7.1 Engine accessories.

3.7.1.1 Starting priming system. If an auxiliary fluid priming system is necessary to meet the requirements of 3.4, one conforming to MIL-E-52649, type

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III shall be furnished. The size of the primer shall be in accordance with the engine manufacturers recommendations. The primer may be mechanically or electromechanically operated. When the primer is mechanically actuated, it shall be furnished with a total cable length not exceeding 6 feet and a "tee" handle. When the primer is electromechanically actuated, it shall be furnished with a temperature control in accordance with MIL-E-52649.

3.7.1.2 Alternator. The alternator shall provide charge at engine low idle speed. The regulator shall be integral with the alternator. Alternator performance shall be in accordance with SAE J180.

3.7.1.3 Starting system. A device shall be furnished to prevent reenergizing the starter motor after the engine has been started. The truck shall be provided with an interlock in the starting system or other means to prevent energizing the starter motor, except when the transmission is in the neutral position. The starter shall be heavy duty and shall be in accordance with the engine manufacturer's recommendations.

3.7.1.4 Fuel system. The capacity of the fuel tank shall provide for not less than 8 hours operation when tested in accordance with 4.5.2.2.18. The tank shall be equipped with a safety filler-cap assembly painted with red enamel which shall incorporate a self-closing cap, screw-locking device for the cap, removable strainer, and shall provide for padlocking. A data plate in accordance with 3.26 or truck marking in accordance with 3.27 shall be furnished and marked "Diesel Fuel" in not less than 1-inch letter height located near the fill pipe. The fuel tank shall be equipped with a drainplug at the lowest point of the tank and be located so that drainage will be into a container without flowing onto any other component of the truck. The drainplug shall be removable with handtools without the removal of any other component.

3.7.1.5 Air cleaner system. A heavy-duty, dry-type replaceable air cleaner shall be furnished. The air cleaner shall be constructed so as to permit disassembly without tools for servicing without removing any other component except the when the engine cowl, hood, or access plate is opened. A restriction indicator shall be furnished.

3.7.1.6 Governor. A speed-limiting type governor shall be furnished. The engine speed shall not exceed the engine manufacturer's recommended intermittent duty speed.

3.7.1.7 Oil filter. A replaceable full flow heavy duty filter(s) with emergency bypass shall be used. The oil filter shall be mounted in an accessible location for ease of replacement of elements.

3.7.1.8 Oil sampling valve. An oil sampling valve in accordance with MIL-V-81940 shall be provided on the engine. The valve shall be located in such a way as to insure that personnel will not be exposed to danger when taking oil samples with the engine running. The location of the oil tap shall be on the filtered oil gallery such that when samples are taken, it shall be a true representation of oil that is flowing while the engine is running. When specified (see 6.2), an oil sampling valve shall be provided for the transmission. Sampling valve shall be labeled crankcase or transmission , whichever is appropriate to indicate the type of oil sampled.

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3.7.1.9 Exhaust system. Muffler and exhaust piping shall conform to the requirements of UL 558. The exhaust system shall be protected against entry of rain. Exhaust gases shall be discharged vertically above the operator's head or at the rear or side of the truck within the plan outline of the truck. The exhaust system shall be arranged to prevent bums to the operator or maintenance personnel while operating or performing maintenance. Guards shall be provided where the placement of the exhaust system is inadequate to prevent burns.

3.8 Transmission. The transmission shall be the full torque, power-shift type in combination with a torque converter. The transmission shall provide the speeds specified on the applicable specification sheet, all of which shall be manually selected. All transmission gears shall be constant mesh. Actuation shall be controlled by a hydraulic-control-valve body mounted on or in the transmission housing. The transmission shall provide for shifting under the full engine power through all successive speeds in each direction. The transmission shall provide controlled application of the clutches to provide smooth shifting without slippage that could damage the clutches. Oil pressure for actuation of the transmission clutches, for torque converter supply, and for transmission lubrication shall be supplied by a pump mechanically driven by the engine. The transmission and torque converter fluid shall be filtered by fullflow filter(s), having replaceable elements. The transmission shall provide for positive inching control or declutch control of the truck (throughout the entire range of engine speed) in the forward and reverse directions. The inching or declutch control shall permit tilting and full lifting speeds with rated load while the transmission control is in forward or reverse position without vehicle movement when tested as specified herein. The inching or declutch control shall apply the service brakes.

3.9 Hydraulic system. The hydraulic system shall consist of all hydraulic components necessary for operation of the mast and forks and does not include brake, transmission, and steering-system components. The system shall provide for lowering of rated load at rate specified in 3.24.2 in the event of failure of or damage to the hydraulic hose supplying the lift cylinder. A pipe tap or straight-thread O-ring boss test point shall be located at the outlet of each pump or pressure inlet of each control valve for measuring pressure. A means to determine fluid level in the reservoir (either dipstick or sight glass) shall be provided. When nuclear safety is specified (see 6.2), the hydraulic lift system shall be fail-safe (see 6.4). A hydraulic-system relief valve shall be provided to protect the hydraulic system from excessive pressure.

3.9.1 System filtration. System filter(s) shall be furnished in the reservoir, the suction line or the return line.

3.9.2 Hydraulic pumps. The hydraulic pump(s) shall have a 1000 hour Omega rating which is less than the system filter Beta Ten rating.

3.9.3 Hydraulic control valves. Mounting of the valves shall be rigid and shall permit no vibration that may cause instability of the controls. All control-valve spools shall be spring centered with sufficient force to return the levers to the center position and retain them there throughout all operations. Spools shall not bind under any operating conditions specified herein. All exposed control-valve surfaces shall show no evidence of corrosion when operated as specified herein.

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3.9.4 Hose reels. When hydraulic hose reels are used on the mast to maintain hose tension during mast extension, sheave wheels, guide rollers, guide blocks, or other means shall be provided to prevent the hose from becoming entangled around and riding over the reels or reel flanges. Hose reels shall be mounted such that they are within the plan outline of the truck for all positions of the forks.

3.10 Uprights and carriage assembly. Uprights shall be of the telescopic, roller type. The drive tires shall not interfere with the carriage assembly travel at any height or upright tilt angle. Chains shall be installed in such a manner that they cannot be displaced from the roller until they have been raised a minimum of 1/2 the chain side plate height. Crossmembers of the mast structure, when in a lowered position, shall not interfere with the operator's horizontal line of sight, ±10 degrees, to the front of the truck. Without leaving the seat, the operator shall be able to see the tip of at least one fork at any lift height when there is no load on the forks. Means shall be provided in the uprights to compensate by replacement or adjustment for wear which may develop between lateral rolling contacts. Positive means shall be provided to prevent overtravel of the carriage or channel in both upper and lower positions. Carriages shall be supplied with adjustable thrust bearings to resist thrust introduced by off center loading. Chains used in the lifting mechanism shall have a factor of safety of not less than 4.7 to 1 based on average ultimate strength. When nuclear safety is specified (see 6.2), load securing rings or eyes shall be provided on both left and right sides of carriage and located not more than 12 inches from top of carriage assembly. Safety factor for load securing devices shall be not less than 2.25 to 1, based on yield strength, assuming that each device is loaded equally. Ring or eye openings shall be determined using MIL-STD-209. The load securing device, when not in use, shall not extend forward of any surface on front face of carriage. The carriage assembly/load securing device combination shall be capable of withstanding a horizontal force (at load securing devices) equal to the truck capacity without failure or permanent deformation of carriage assembly. Direction of force is forward. The mast shall have two lift cylinders, each located in the middle of the mast, except a free lift cylinder no higher than the top of the fork carriage is acceptable. Any hydraulic hoses or chains located within this open area shall be located away from the center of the mast.

3.10.1 Forks. Forks shall conform to ASME B56.1 and the applicable specification sheet. The fork spacing detailed on the specification sheets shall be measured from the outside to outside surfaces of the forks. Fork thickness shall not exceed 1/2 inch at the tips and the end taper shall be not less than 14 inches long. Taper shall be on the underside of the forks. The top of the fork tips shall be chamfered or rounded to break corners and provide smooth edges.

3.10.2 Fork positioner. Hydraulic powered fork positioners shall be furnished. Fork spacing shall be simultaneously hydraulically adjustable within the dimensions specified on the applicable specification sheet. The dimensions shall be measured from the outer edges of the forks.

3.10.3 Side shift. The truck shall be equipped with a hydraulically powered side-shift mechanism. The side-shift mechanism shall side shift rated load a minimum of 6 inches each side of center (12 inches total travel).

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3.10.4 Load backrest. A metal load backrest, removable without disturbing any other component, shall be provided. There shall be no protruding bolts or appendages beyond the side plane of the load backrest. The load backrest in conjunction with fork and carriage design shall provide a vertical rear guard at least 48 inches high measured from load carrying surfaces of the forks. It shall have width and capacity and strength sufficient to prevent a rated load 48 inches high or less from forcing the backrest against the upright when the mast is at maximum rearward tilt. The backrest shall be capable of withstanding without permanent deformation a horizontal static load equal to the rated load uniformly distributed over the area, from the top carrying surface of the forks to the top of the backrest for not less than 3 minutes.

3.11 Steering. An automotive type steering wheel shall be furnished. There shall be a minimum of 7 inches measured vertically between the lowest point on the steering wheel rim and a line parallel to the floor and passing through the highest point of the unoccupied seat cushion. There shall be a minimum of 3 inches between the rim of the steering wheel and the directional control lever(s) for all positions of the lever(s). Ackermann steering shall be applied. Clock-wise rotation of the steering wheel shall provide for right-hand turning of the truck while in forward motion. Provision shall be made for adjustment for wear and alignment of all major steering components and for lubrication, if required, of all friction points by accessible grease fittings, packing the bearings or permanent lubrication. The truck shall negotiate the obstacle course in the reliability test with only the displaced wheel off the ground at any time. Means shall be provided to prevent wheels or tires or steering linkage from contacting any part of truck structure, other than stops, irrespective of obstacle magnitude or steer angle. The steer wheels shall be capable of being turned from extreme right to extreme left or vice versa in not more than six complete turns of the hand steering wheel. Power steering shall be furnished and shall conform to SAE J53. In the event of power steering pump failure or engine failure, the forces required to steer the vehicle shall not exceed the limits specified in SAE J53.

3.12 Brakes.

3.12.1 Service brake system. The brake system shall conform to ANSI B56.1. Power brakes shall be provided and shall charge the system to maximum power setting within 2 minutes when the brake actuating energy is fully exhausted. If hydraulic brakes are utilized, the brake system fluid shall be in accordance with MIL-B-46176. If air brakes are furnished, an air drying system with provisions to drain water from the air reservoir(s) shall be provided in the air system. An audible visual low power warning device shall also be provided with an air brake system. Automatic or manual brake adjustment shall be provided. Automatic adjustment shall maintain correct lining to drum or disc adjustment without looseness or lockup. The automatic adjusters shall be of the guarded or enclosed type, protected from environmental damage or malfunction. In the event of power assisted failure, the force the operator must exert on the brake pedal to hold the truck stationary on a 20 percent grade under all loading conditions shall not exceed 70 pounds.

3.12.2 Parking brake. Truck shall be equipped with an independent mechanical friction parking brake mounted on the drive or propeller shaft

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rotating with the wheels. The parking brake shall be hand or foot operated. In lieu of friction brake, the service brakes may be utilized as a parking brake. However, activation of parking brakes shall be independent of the service brake hydraulic system and shall employ a mechanical activation system. The parking brake shall conform to all requirements of ANSI B56.1, except the brake shall hold the truck stationary on a 20 percent grade. The force that the operator must exert on the parking brake control to achieve the above listed brake requirements shall not exceed 30 pounds for hand operated parking brakes and shall not exceed 70 pounds for foot operated parking brakes.

3.12.2.1 Parking and emergency stopping system. When nuclear safety is specified (see 6.2) a parking and emergency stopping system shall be provided which shall be an isolated system actuating the service brakes in accordance with SAE J1152. The emergency stopping system shall be capable of stopping the loaded truck at maximum speeds when demonstrated in accordance with 4.5.2.2.8.2. The parking brake shall be capable of holding the truck on a grade of 20 percent with or without rated load. The criterion shall apply to both forward and reverse directions.

3.13 Truck access. Operator access to the truck shall be in accordance with minimum requirements of SAE J185.

3.14 Seat. A cushion seat and backrest shall be furnished in accordance with the minimum requirements specified in SAE J899. The seat location shall conform to SAE J898.

3.15 Cowl or hood. The engine shall be completely enclosed with a metal hood cover and side plates. Hood cover shall be capable of being opened or removed by one person. Handtools shall not be required for release of hood or side plates.

3.16 Towing device. A pintle or pin-type towing device shall be provided in the center rear of the truck. The towing device shall accept lunette conforming to MS51336 without binding. The towing device shall not protrude beyond the path followed by the rear end of the truck when making a turn of minimum turning radius. The towing device shall be capable of withstanding, without physical failure, a towing force induced by the drawbar drag as specified in ANSI B56.1.

3.17 Batteries and battery cables.

3.17.1 Batteries. The truck shall be furnished with a minimum of one 12-volt battery conforming to SAE J537. The battery shall be charged and dry.

3.17.2 Battery cable terminals. Battery cables shall be in accordance with SAE J541. The positive cable shall be colored red and labeled “+”. The negative cable shall be colored black and labeled “-”.

3.18 Instrument panel. The instrument panel shall be located so that the operator can read the instruments from the normal operating position. The instrument panel shall be installed on an inclined or vertical plane to facilitate drainage of water. On this panel shall be mounted the electrical system monitor, engine oil pressure gage or indicator light, engine coolant temperature gage or indicator light, transmission fluid temperature gage or

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indicator light, fuel gage, engine run control, light switches, starter switch, airbrake pressure gage (if applicable), and any other instruments used. The engine run control and the starter switch may be provided on the steering column if that is standard for the commercial product. All gages, indicator lights, and switches shall be identified on the face of the gage in accordance with SAE J1362 or as specified in 3.26. All gages shall be mounted flush and internally lighted.

3.19 Controls and instrumentation. Each control shall be provided with a position marking diagram on a plate as specified in 3.26, or embossed or depressed on or near the control. Markings for levers which control the lift, mechanism shall conform to ASME B 56.11.3. The diagrams shall be visible to the operator from a normal operating position. All controls for the truck shall be located in the operator's compartment. Electric controls and instrumentation shall be moisture and weather resistant. Mechanical push-pull remote controls, when furnished, shall be of the waterproof antifriction type, conforming to requirements for marine cables specified in SAE J917.

3.19.1 Light switches. A minimum of two light switches shall be furnished on the instrument panel. One switch shall control the forward floodlight(s) as specified in 3.20.1 and taillight as specified in 3.20.2. One switch shall control the rear floodlight(s) as specified in 3.20.1.

3.19.2 Engine run control. An engine run control shall be furnished and shall be a key type. It shall have not less than two positions "on" and "off". The switch shall control the operation of monitors and of all electrical equipment except the horn. Two keys shall be furnished with each truck. A mechanical engine shutoff ray be furnished and shall be in addition to the emergency shutoff as specified. All keys shall be identical for every truck.

3.19.3 Electrical system monitor. A remote indicating ammeter or alternator indicator red light shall be furnished. The ammeter shall be of sufficient capacity to indicate variation in current supply or demand experienced by the truck, exclusive of starter requirements. The indicator light shall light when the alternator is not charging and the engine run-control is "on".

3.19.4 Engine oil pressure monitor. A remote indicating engine oil pressure gage and a low-pressure red indicator light shall be furnished. The indicator light shall be provided with a means to check operation of the light.

3.19.5 Engine coolant temperature monitor. A remote indicating engine coolant temperature gage or a high-temperature red indicator light shall be furnished. If an indicator light is furnished, means shall be provided to check operation of the light. The light shall come "on" when the engine coolant temperature exceeds the temperature specified by the engine manufacturer.

3.19.6 Transmission temperature monitor. A remote indicating transmission temperature gage and a high-temperature red indicator light shall be furnished. Means shall be provided to check operation of the light. The light shall come "on" when the transmission temperature exceeds the temperature specified by the transmission manufacturer.

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3.19.7 Hour meter. An electrically operated hour meter, which registers the number of engine operating hours only, shall be provided. The hour meter shall be capable of totaling a minimum of 9999 engine hours. The hour meter shall be accessible for reading without removal of any component of the truck.

3.19.8 Fuel gage. An electric, thermally stabilized, remote indicating fuel gage shall be provided.

3.20 Electrical system. The truck shall have a nominal 12-volt electrical system. Multiple conductor routing shall be protected by flexible tubing or plastic electrical insulation tape. Grommets or suitable fasteners shall be provided whenever wiring passes through bulkheads, partitions or structural members; and the wiring shall be fastened to prevent chafing or abrasion. All wiring shall be color or number coded, and the coding scheme shall be included on the electrical diagrams. All wires shall be adequately sized, and all circuits except the ignition shall have overload protection. All electrical connections fastened with terminal nuts shall be equipped with lock washers or nut locking devices.

3.20.1. Lights. Each truck shall be equipped with a minimum of four sealed beam floodlights conforming to SAE J598, PAR 46, Trade No. 4478, which shall be shock mounted in elastomer ring housings, two in the front, two in the rear, with the two front floodlights mounted on the outside of the uprights (one each side). Each light shall be adjustable a minimum of 45 degrees above and below horizontal plane. Adequate protection against damage of each light, by means of position or guard, shall be provided.

3.20.2 Taillight. One automotive type combination stoplight and taillight shall be furnished. The taillight shall conform, to SAE J585 and J586. The lamps shall be either sealed units mounted in elastomer rings or bulbs and lens mounted in elastomer housings. The bulb shall be Trade No. 1157. The taillight shall conform to SAE J759, red minimum 3-inch diameter, and shall be recessed or mounted on the rear and within the plan outline of the truck, and shall be protected against damage by a guard if not recessed.

3.20.3 Horn. The truck shall be equipped with an electric horn with a pushbutton mounted in the center of the steering wheel. The horn button assembly and electrical wiring for the horn shall be constructed to be moisture and weather resistant to prevent entry of moisture when operated or stored outdoors under all weather conditions.

3.20.4 Warning light. When specified (see 6.2), the truck shall be equipped with a flashing amber light visible from all directions, which operates continuously when activated by an "on-off" switch located on the instrument panel adjacent to the light switches specified herein. The light shall conform to SAE J96 or SAE J845.

3.21 Operator's overhead guard. An overhead guard shall be furnished and shall be in accordance with ANSI B56.1.

3.22 Personnel cab. When specified (see 6.2), the truck shall be equipped with an insulated, weathertight personnel cab. The cab clearances shall be in accordance with SAE J154. The cab shall inherently resist torsional strain and withstand all operational shock and vibration under the operating conditions specified herein. The cab shall be constructed of metal. The cab

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shall be equipped with a minimum of two fully opening or removable doors. Each door shall have a safety latch, opening stop and hold-open catch. All hardware shall be of the heavy-duty or industrial type, resistant to corrosion. The cab shall be equipped with not less than two ventilation fans not less than 6 inches in diameter. Fans shall be directionally adjustable to direct airflow toward all glazing. Fan(s) shall have "on-off" switch(es) which are labeled “ON” and "OFF" and are within the seated functional reach of the 5th percentile female operator. The cab shall be removeable and independent from the overhead guard specified herein.

3.22.1 Glazing. The cab shall provide the operator with all around visibility. The total glazed area shall be not less than 4000 square inches of tinted windows consisting of windshield, sliding or roll-down door windows, rear window, roof window, and other windows as determined by cab design. All glazing shall conform to ANSI Z26.1, type AS1 in the windshield and roof window and type AS1 or AS2 in other windows. In lieu of type AS1 glazing for the roof window, shock resistant clear plastic treated to prevent scratches due to window wiper operation and which shall not become opaque throughout the life of the truck can be used. Fixed glazing shall be sealed in waterproof stress-absorbent molding, and movable windows shall be sealed on extra heavy type, pile lined channels, or heavy duty self-draining plastic channels.

3.22.2 Interior. The interior of the cab shall be fitted with insulation, retarding both heat loss and noise intrusion. The insulation and framing shall be covered with an interior skin and headliner, providing a smooth, cleanable protective cover.

3.22.3 Windshield and window wipers. The cab shall be equipped with at least three electrically powered-wiper assemblies to clear the windshield, the rear window and the roof window in accordance with the performance requirements of SAE J198. Self-parking features shall be provided. Wiper arms and blades shall be of the commercial heavy-duty type having saddle and pin blade fastenings. The wiper arms shall provide tension on the glazed surface of at least 1 ounce for each inch of wiper arm length. The operator controls shall be installed on or adjacent to the instrument panel.

3.22.4 Heater. A recirculating-air type heater shall be furnished. The heater shall have a rating of at least 18,000 Btu/hour. The heater controls shall be near the operator and shall provide for variable selection of blower speeds.

3.23 Lubrication fittings. Lubrication fittings shall conform to SM J534. Fittings shall be located in a protected accessible position. Fittings shall be accessible by all grease guns conforming to MIL-G-3859, equipped with a hydraulic coupler conforming to MS24207 and a 10-inch flexible extension. Accessibility or fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand operated fasteners may be removed to provide accessibility. Accessibility shall be in accordance with 3.6.

3.24 Performance.

3.24.1 Lifting speed. Speed of lift with rated load shall be not less than 55 feet per minute over the entire distance from ground level to maximum lift height.

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3.24.2 Lowering speed. Speed of lowering forks with rated load shall be not more than 80 feet per minute and not less than 40 feet per minute at engine idle speed over the entire distance from maximum lift height to ground level. The hydraulic control system shall be damped, metered or manually regulated to reduce shock and prevent the truck from overturning when rated load is lowered at maximum speed and then stopped quickly. When nuclear safety is specified (see 6.2), maximum lowering speed, immediately before forks contact ground, shall be not more than 65 feet per minute.

3.24.3 Right angle turn. The truck, when carrying rated capacity load, shall be capable of backing through a turn in either direction and returning to the same or a parallel position within dimension "A" of figure A-1 when demonstrated as specified herein.

3.24.4 Speed. While carrying the rated load on a level surface, the truck shall be capable of attaining a minimum speed of 12 miles per hour in both the forward and reverse directions.

3.24.5 Slope ascension, first speed, forward direction. The truck shall be capable of ascending a 20 percent grade on a dry concrete or asphalt surface with or without rated capacity load. A truck carrying the rated load shall be able to accelerate from a dead stop on this slope.

3.24.6 Upright tilt. When unloaded, truck masts shall have 3 degrees, plus 1/2 minus 0 degrees, forward tilt, and 10 degrees, ±1/2 degree, rearward tilt. Positive means shall be furnished to prevent cavitation of the tilt cylinders at minimum 120 °F oil temperature when tilting rated load to full forward tilt at engine law idle speed. When nuclear safety is specified (see 6.2), the hydraulic tilt circuit shall be provided with a flow restrictor or locking valve arrangement to prevent uncontrolled tilting in the event of a hydraulic line failure.

3.24.7 Upright height. Collapsed vast height of the unloaded truck measured from the ground to the top of the uprights in a vertical position shall not exceed that specified in the applicable specification sheet.

3.24.8 Maximum fork height. The maximum fork height of a truck bearing rated capacity load and the uprights fully extended, measured from the ground to the load carrying surface of the fork at the rated load center, shall be not less than that specified in the applicable specification sheet.

3.24.9 Free lift height. Height of forks without increase in the specified collapsed mast height shall be not less than that specified in the applicable specification sheet.

3.24.10 Underclearance. Minimum underclearance beneath the mast assembly, when in true vertical position, shall be not less than that specified on the applicable specification sheet when the truck is loaded. When the truck is unloaded, the underclearance of the truck other than the mast and axles, shall be not less than 7 inches. Hydraulic fittings, hoses, tubing, and linkages shall not be the lowest portion of the truck and shall be protected by structural members from striking obstacles.

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3.24.11 Drift. The lift assembly shall be capable of holding the rated load at maximum height not less than 2 minutes with not more than 1-3/4-inch of vertical drift and not more than 1 degree of rotational drift from the vertical.

3.24.12 Stability. The truck shall meet "forward stacking", "forward travel", "lateral stacking" and "lateral travel" stability requirements of ANSI B56.1.

3.24.13 Truck operations. Unless otherwise specified (see 6.2), the truck shall complete 6 to 8 circuits per hour for a cumulative total of 100 hours, when demonstrated as specified in 4.5.2.2.18. The 100 hour test shall be conducted on a continuous basis during each operating 8 hour day (minimum) and 40 hours per week (minimum). When specified (see 6.2), the cumulative total shall be 24 hours conducted non-stop for the total hours. However, truck shall complete either the 24 or 100 hours without any failures. If failure(s) occur before the completion of the required hours, the failure(s) shall be corrected and the test restarted beginning with the first hour. Failure for the purpose of this test is defined as any malfunction(s) which cannot be remedied within 15 minutes by adjustment, repair, or replacement action by contractor maintenance personnel using contractor's tools and parts and which may cause:

a. Failure to commence operation, cessation of operation or degradation of performance below designated levels.
b. Damage to truck by continued operation due to fluid leaks, overheating, or internal component failure of the engine, transmission, axles, brakes, differential and hydraulic pump.
c. Serious personnel safety hazard.
d. The total time of all malfunctions exceeds 2 percent of the cumulative total hours.
e. Truck overturn which results in either the steer or drive wheel losing contact with the ground more than 5 seconds.

Only routine maintenance as prescribed by the technical manual is allowed. Record the reason and amount of time involved for *each* truck stoppage or malfunction, including refueling, during the truck operation test.

3.24.14 Noise limits. The noise produced by the truck shall conform with MIL-STD-1474 requirements (with the exception of 5.2, 5.3 and 5.4) when tested in accordance with specification (see 4.5.2.2.19). The provisions of MIL-STD-1474, 4.3 and 4.4 shall be provided if and only if MIL-STD-1474, 5.1.1.2 procedures have been pursued and documented to the satisfaction of the contracting officer and written permission to exceed the 85 dB(A) limit is obtained from the contracting officer. Hazard signs shall conform with MIL-STD-1474, 4.3 and be readable at the distance indicated on the sign.

3.25 Treatment and painting. The parts of the truck to be painted shall be cleaned, treated and painted in accordance with the contractor's commercial practice. Unless otherwise specified, final color shall be contractor's commercial yellow. When specified (see 6.2) the portions of the truck normally painted, both internally and externally, shall be cleaned, treated, and painted in accordance with MIL-STD-704, type F for ferrous metals and type G for nonferrous metals and fiberglass top coat camouflage green 383. Primer in accordance with MIL-P-53030 may be used. Finish coat accordance with MIL-

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C-53039 (polyurethane) may be used in lieu of MIL-C-46168. Vendor part and components received by the prime contractor with either alkyd primer or alkyd paint coating may be coated with above specified primers and paints applied directly over existing alkyd primer or paint without additional service preparation except for cleaning, if the base alkyd primer or paint meets applicable military federal specifications and if the base alkyd primer or paint is allowed to cure for a period of 14 days. This exception for vendor parts and components is not intended to supercede the requirement that the complete truck be primed and painted in accordance with MIL-T-704 using the primers and paints specified above. If another color is specified (see 6.2), the above system shall be overcoated with MIL-C-46168 or MIL-C-53039 of the desired color.

3.25.1 Walkway coating. Floor plates and step surfaces shall be coated with a nonslip coating compound or be furnished with a nonslip metal surface.

3.26 Identification marking. All identification, instruction and marking plates shall conform to MIL-P-514, of type I, style 1, Composition C (GG-P-455, type I, grade A, class 1) material. All plates shall be securely attached by screws, bolts or rivets in a conspicuous location, and shall be furnished and mounted by the contractor. The truck shall be identified with an identification plate located in a visible and safe location. The identifying data to be applied to the identification plates shall be as indicated herein as defined in MIL-STD-130. The truck shall also be equipped with instruction, warning, and caution plates conforming to MIL-P-514, type III, composition C (GG-P-455, type I, grade A, class 1), and describing any special or important procedure to be followed in assembling, operating, loading and servicing of the truck and its components.

3.26.1 Shipping data plate. A shipping data plate conforming to MIL-P-514, type III composition C (GG-P-455, type I, grade A, class 1), shall be furnished and in addition shall show the silhoutte of the truck in transport position indicating the center of gravity. Wheel loading information specified in 3.26.2 may be combined in one plate with the requirements specified herein.

3.26.2 Wheel loading plate. Each truck shall be equipped with a wheel loading plate conforming to MIL-P-514, type III, composition C (GG-P-455, type 1, grade A, class 1). As a minimum, the plate shall have the following information:

Wheel loading (no load on forks)
Drive wheels (each) pounds
Steer wheels (each) pounds

Wheel loading (gross vehicle weight)
Drive wheels (each) pounds
Steer wheels (each) pounds

3.26.3 Safety rating. When type DS truck is specified (see 6.2), each truck shall be equipped with a safety designation plate conforming to MIL-P-514 type III, composition C (GG-P-455, type 1, grade A, class 1). As a minimum, the plate shall indicate truck manufacturer's name, truck's complete model number, safety designation (type DS), independent testing laboratory's

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name, independent testing laboratory's registration or index number assigned to the inspected truck and data and location of inspection.

3.27 Truck marking. Markings on truck shall be painted lusterless black. Characters shall be block-type capitals and Arabic numerals. Location, information, and character size shall be as shown in table I.

TABLE I. Truck marking.

Information	Example	Location	Character Size (Height)
Agency US		On each side and rear	1-1/2 inches
Registration number	1OD315	On each side and rear	1-1/2 inches
Capacity	20,000 – CAP	On each side of mast	2 inches
Safety	No smoking No riders	On rear of mast	2 inches
Tire pressure	TPXX	on frame above each tire	1 inch
Fuel fill rate	Max. safe fuel accept - gpm	Near fuel tank filler	1 inch

3.28 Tires and tire loadings. Tire loadings for the selected size shall not exceed the values at speeds specified herein listed in the Tire and Rim Yearbook (as applicable). Tires shall be not more than 12 months old on date of acceptance by the Government. The maximum tire pressure permitted shall be 125 psi.

3.29 Electromagnetic radiation. Unless otherwise specified (see 6.2) the forklift shall not exceed the limits of electromagnetic radiation specified in SAE J551. Measurement shall also be taken from the front and rear of the truck at 33 foot distances.

3.30 Technical information package (TIP). When specified (see 6.2), the manufacturer shall complete the TIP-52864 to depict the truck and its components as it will be furnished.

3.31 Fenders. Fenders shall be provided for all exposed wheels in accordance with the manufacturer’s standard commercial practices.

3.32 Transportability. The trucks shall be capable of being transported by military or commercial trailers, trains, and marine vessels, and must be capable of withstanding the impact forces encountered in shipment, without damage or permanent deformation. When specified (see 6.2), the trucks shall be equipped with tiedown or slinging provisions.

3.32.1 Tiedown Provisions. The tiedown provisions shall conform to MIL-STD-209, class 2 or 3, type III equipment. The tiedown provisions shall satisfactorily complete the pull testing as specified without weld failure, permanent deformation, cracking, loosening, or breaking of the provision or its connecting structural components.

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3.32.2 Slinging Provisions. The slinging provisions shall conform to MIL-STD-209, class 1 or 3, type III. The provisions shall enable the complete trucks to be lifted in *the* normal operating position. The provisions shall be located so that not less than one inch clearance is maintained between slings and all exterior parts and shall be fastened to members which will withstand stresses in the amount and direction of pull specified for the provisions without weld failure, permanent deformation, cracking, loosening, or breaking of the provisions or its connecting structural components. Slinging provisions may also be used as tiedown provisions when such provisions meet the requirements specified in 3.32.1. All slinging/ tiedown provisions shall be labeled 'LIFT','TIEDOWN', or 'LIFT TIEDOWN' as appropriate, in 1-inch (2 .54 cm) high letters.

3.32.3 Rail transportability. The trucks shall be rail transportable in CONUS and NATO countries without restrictions. The trucks shall have a dimensional profile within the Gabarit International de Chargement (GIC) in accordance with MIMC/TEA pamphlet 70-1, outline diagram when loaded on a 50-inch (127 cm) high rail car. The trucks shall be capable of withstanding shock loads resulting from rail impact testing in accordance with 4.5.2.2.23.1.2.3 without failure, damage, or permanent deformation.

3.32.4 Highway transportability. The trucks when loaded on a semitrailer/ tractor, shall be within the highway permit limits of all States.

3.32.5 Water transportability. The truck shall be transportable on the LARCLX and large vessels/ships.

3.33 Maintainability. Provision shall be made for checking all fluid levels, adjustment, servicing, and replacement of all electrical assemblies and components, hydraulic system components, battery, wearing parts of steering assembly, tires, wheels, lights and horn. All fluid levels shall be accessible when the engine is at operating temperature without subjecting personnel to injuries or burns. If hand access openings are required to perform maintenance, the edges of each opening shall be smooth and the opening shall be provided with a removable or hinged cover. Dimensions of access openings shall be in accordance with SAE J925 for mittened hand. There shall be no interference with the servicing or drawing of oils and lubricants to or from any assembly or component by frame members or other obstructions. Oil and lubricants shall be capable of being drained without requiring the use of flexible connections. All drains shall provide for complete drainage. Engine, transmission, hydraulic system, steering system, and service brakes shall be furnished with dipsticks in accordance with SAE J614. In lieu of dipstick to check service brake and hydraulic system visual inspection will be acceptable.

3.33.1 Maintenance operations. Unless otherwise specified herein, each operation, in the following maintenance operations list, shall be accomplished by one person in not more than the time specified using common tools or any special tool, if furnished with the truck:

a. Remove replace and adjust all engine driven belts - 1/2 hour.
b. Remove and replace alternator - 1/2 hour.
c. Remove and replace all filters, screens and strainers in all hydraulic systems - 1/2 hour.
d. Remove and replace fuel filter elements - 1/2 hour.

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e. Drain engine lubricating oil, remove and replace oil filter elements and refill crankcase - 1/2 hour.
f. Remove and replace engine coolant system hoses - 1/2 hour.
g. Disconnect battery cables, remove and replace batteries and reconnect battery cables - 1.2 hour.
h. Drain torque convertor oil and transmission oil, remove and replace all filter elements and strainers and refill convertor and transmission - 1 hour.
i. Remove and replace starter - 1 hour.
j. Bleed and adjust brakes and refill master cylinder (two persons) - 3/4 hour.
k. Remove and replace headlight, floodlights and taillight bulbs - 1/4 hour each bulb.
l. Remove, replace and reinstall air filter - 1/2 hour.
m. Lubricate all lubrication fittings with four strokes per fitting using grease gun and extension if required - 1 hour.
n. Remove, read and replace each dipstick or conduct visual inspection - 2 minutes.
o. Drain water separator or replace filter element - 1/2 hour.

3.34 Endurance/reliability. The truck shall have a mean time between failure (MTBF) of 75 hours when tested in accordance with paragraph 4.5.2.2.24.

3.35 M16 mounting. Unless otherwise specified (see 6.2), a mounting bracket, for securing an M16 rifle, conforming to TACOM Drawing number 12276986 shall be provided and mounted in a suitable location or near the operators compartment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of Packaging (see 4.6).

4.3 First article inspection.

4.3.1 First article examination. The first article shall be examined as specified in 4.5.1. The presence of one or more defects shall be cause for rejection.

4.3.2 First article test. The first article shall be tested as specified in 4.5.2. Failure of any test, permanent deformation, over-heating, malfunction, failure to meet performance requirements, failure to complete any test, leakage of fuel, leakage of hydraulic oil, or lubricants shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIA. A lot shall be accepted when 0 defects are found and rejected when 1 or more defects are found.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1 unless otherwise specified (see 6.2). When specified (see 6.2), each truck in the lot shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.3 Tests.

4.4.3.1 Individuals. Each truck shall be subjected to the tests marked “X” in column 2 of table III. Failure of any test shall be cause for rejection.

4.4.3.2 Samples. Samples selected in accordance with 4.4.1 shall be subjected to the tests marked “X” in column 3 of table III unless otherwise specified in 6.2. Failure of any tests shall be cause for rejection.

4.5 Inspection procedure. Prior to examination or tests, the truck shall be lubricated with military or commercial oils and greases for use at the ambient temperature at the inspection site.

4.5.1 Examination. The truck shall be examined as specified herein for the defects listed in table II.

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TABLE II. Examination schedule.

Number	Defect	Requirement paragraph
101.	Materials not as specified.	3.3
102.	Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.	3.3.1
103.	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.3.2
104.	Contractor does not have documentation available for identification of material, material finishes, or treatment.	3.3.3
105.	Used, rebuilt or remanufactured components, pieces, or parts incorporated in the fork lift truck.	3.3.4
106.	Safety not as specified.	3.5
107.	Nuclear safety not as specified.	3.5.1
108.	Fire extinguisher not as specified.	3.5.2
109.	Rearview mirror not as specified.	3.5.3
110.	Human factors engineering not as specified.	3.6
111.	Engine not as specified.	3.7
112.	Starting priming system not as specified.	3.7.1.1
113.	Alternator not as specified.	3.7.1.2
114.	Fuel system not as specified.	3.7.1.4
115.	Air cleaner system not as specified.	3.7.1.5
116.	Governor not as specified.	3.7.1.6
117.	Oil filter not as specified.	3.7.1.7
118.	Oil sampling valve not as specified.	3.7.1.8
119.	Exhaust system not as specified.	3.7.1.9
120.	Hydraulic system not as specified.	3.9
121.	System filtration not as specified.	3.9.1
122.	Hydraulic pump not as specified.	3.9.2
123.	Hydraulic control valves not as specified.	3.9.3
124.	Hose reels not as specified.	3.9.4
125.	Forks not as specified.	3.10.1
126.	Fork positioner not as specified.	3.10.2
127.	Load backrest not as specified.	3.10.4
128.	Truck access not as specified.	3.13
129.	Seat not as specified.	3.14
130.	Cowl or hood not as specified.	3.15
131.	Towing device not as specified.	3.16
132.	Batteries and battery cable terminals not as specified.	3.17
133.	Instrument panel not as specified.	3.18
134.	Controls and instrumentation not as specified.	3.19
135.	Electrical system not as specified.	3.20
136.	Lights not as specified.	3.20.1
137.	Taillight not as specified.	3.20.2
138.	Horn not as specified.	3.20.3
139.	Warning light not as specified.	3.20.4
140.	Operator's overhead guard not as specified.	3.21
141.	Personnel cab not as specified.	3.22

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TABLE II. Examination schedule (Con't)

Number	Defect	Requirement paragraph
142.	Glazing not as specified.	3.22.1
143.	Interior not as specified.	3.22.2
144.	Windshield and window wipers not as specified.	3.22.3
145.	Heater not as specified.	3.22.4
146.	Lubrication fittings not as specified.	3.23
147.	Treatment and painting not as specified.	3.25
148.	Walkway coating not as specified.	3.25.1
149.	Identification marking not as specified.	3.26
150.	Shipping data plate not as specified.	3.26.1
151.	Wheel loading plate not as specified.	3.26.2
152.	Safety rating plate not as specified.	3.26.3
153.	Truck marking not as specified.	3.27
154.	Technical information package not as specified.	3.30
155.	Fenders not as specified.	3.31
156.	M16 mounting bracket not as specified.	3.35

4.5.2. Tests. The truck shall be tested to demonstrate compliance with the requirements set forth in section 3 in accordance with procedures shown in section 4. When the contractor can provide verifiable data demonstrating to any or all parts of the specified testing, these tests may be waived by the Government. Failure to complete the requirements of one or more tests shall constitute failure of the tests. Table III indicates the items that shall be tested and the paragraphs which apply.

4.5.2.1 Test schedule.

4.5.2.2 Test methods. Tests shall be conducted in accordance with appendix tests methods, and as specified herein. Unless otherwise specified herein, tests shall be conducted at the ambient temperature and climatic conditions. Only that maintenance established by the contractor and submitted as a maintenance schedule prior to commencement of the test shall be performed during the test. Test may be conducted in any order except the truck operations test shall be conducted last.

4.5.2.2.1 Rated load. The tolerances for weight shall be +0 percent and -1 percent and for dimensions shall be ±1 percent.

4.5.2.2.2 Environmental requirements. Demonstrate that the truck can start without preheating within 5 minutes and operate as specified within 15 minutes after engine start as specified in 3.4. The forklift shall have been cold soaked to attain a stabilized temperature specified. Nonconformance shall constitute failure of this test.

4.5.2.2.3 Human factors engineering. The human factors engineering characteristics of the truck will be evaluated during all testing. Nonconformance to 3.6 shall constitute failure of this evaluation.

4.5.2.2.4 Starter disconnect. Place the transmission in gear and engage the starter switch. Repeat for all forward and reverse gears. Any evidence

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of starter engagement shall constitute failure of this test. Start the truck by energizing the starter switch. Release starter switch and immediately reactivate starter switch while engine is operating. Any evidence of starter engagement while engine is running shall constitute failure of this test.

4.5.2.2.5 Production run-in. Each truck carrying a rated load, shall be subjected to a production run-in and servicing as specified herein. Inability to complete the run-in, overheating, failure of any components, malfunction of any control, or evidence of leakage of fluids shall be cause for rejection of the truck by the Government until such deficiencies are corrected.

4.5.2.2.5.1 Travel. Demonstrate starting, ascending and descending a 20 percent ramp, steering, inching and high speed travel. Operate all lights, horn, and brake system not less than two times during this test.

4.5.2.2.5.2 Transmission. Demonstrate that the number of speeds forward and reverse specified on the applicable specification sheet are provided. Demonstrate shifting under full power through all successive gear ratios in each direction. Demonstrate positive inching control in each direction and show that this control can declutch the transmission and brake the truck in accordance with 3.8.

4.5.2.2.5.3 Mast assembly. Demonstrate ability to lift rated load to full lift height at full lift speed and lower load at full opened valve without exceeding maximum lowering speed. Demonstrate ability to tilt mast full forward and full rearward with rated load 12 inches off the ground. Demonstrate ability to side shift rated load throughout full range of side shift. Demonstrate ability to feather hydraulic for control of all hydraulic functions. Each operation shall be performed at least three times.

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TABLE III Test schedule.

First Article	Individual	Sample	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6
X			Environavental requirements	4.5.2.2.2	3.4
X		X	Human factors engineering	4.5.2.2.3	3.6
X		X	Starter disconnect	4.5.2.2.4	3.7.1.3
X	X	X	Production run-in	4.5.2.2.5	3.8, 3.11, 3.12.1 3.19.1, 3.20.3, 3.24.4, 3.24.5
X		X	Load securing devices and carriage assembly	4.5.2.2.6	3.10
X		X	Side shift	4.5.2.2.7	3.10.3
X			Service brake system	4.5.2.2.8.1	3.12.1
X			Parking brakes	4.5.2.2.8.3	3.12.2
X		X	Lifting and lowering speed	4.5.2.2.9	3.24.1 3.24.2
X	X	X	Right angle turn	4.5.2.2.10	3.24.3
X	X	X	Travel speed	4.5.2.2.11	3.24.4
X	X	X	Slope ascension	4.5.2.2.12	3.24.5
X		X	Tilt	4.5.2.2.13	3.24.6
X		X	Upright height, maximum fork height and free lift height	4.5.2.2.14	3.24.7 3.24.8 3.24.9
X	X		Underclearance	4.5.2.2.15	3.24.10
X	X	X	Drift	4.5.2.2.16	3.24.11
X			Stability	4.5.2.2.17	3.24.12
X		X	Truck operations	4.5.2.2.18	3.24.13
X			Noise level	4.5.2.2.19	3.24.14
X			Tire loading	4.5.2.2.20	3.28
X			Electromagnetic radiation	4.5.2.2.21	3.29
X			Emergency stopping system	4.5.2.2.8.2	3.12.2.1
X			Maintenance operations	4.5.2.2.22	3.33.1
X			Endurance/reliability testing	4.5.2.2.24	3.34
X			Transportability provisions	4.5.2.2.23.1.2 thru 4.5.2.2.23.1.2.5	3.32 thru 3.32.5
X			Operation with MIL-T-83133	4.5.2.2.25	3.7

4.5.2.2.5.4 Servicing and adjustment. Unless otherwise specified (6.2), prior to delivery, the manufacturer shall service and adjust each truck for customer operational use, including at least the following:

a. Adjustment of mast assembly.
b. Adjustment of engine, transmission, and power train.
c. Adjustment of brake system.

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d. Adjustment of steering.
e. Continuity check of the electrical system, and charging of battery.
f. Check torque on all wheel lug nuts and proper inflation of tires.

4.5.2.2.6 Load securing devices and carriage assembly. When load securing devices are provided, the devices and carriage assembly shall be tested for verification of specified requirements. Nonconformance to 3.10 shall constitute failure of this test.

4.5.2.2.7 Side shift. Operate the side shift attachment for 100 full cycles at full speed from extreme left to extreme right and back to extreme left with the rated load. During the last 50 cycles, the control shall be released abruptly when the load is in approximately midposition between the left and right extremes and then the side shifter shall be restarted. Nonconformance to 3.10.3 shall constitute failure of this test.

4.5.2.2.8 Brakes.

4.5.2.2.8.1 Service brake system. Perform applicable braking test specified in ASME B 56.1. Nonconformance to 3.12.1 and ASME B 56.1 shall constitute failure of this test.

4.5.2.2.8.2 Emergency stopping system. When an emergency stopping system for nuclear safety as specified in 3.12.2.1 is provided, stopping distance shall be demonstrated on a level surface in accordance with ASME B 56.1. Nonconformance to 3.12.2.1, ASME B 56.1 and maximum stopping distances shown in SAE J1152 shall constitute failure of this test.

4.5.2.2.8.3 Parking brake system. Perform applicable braking test specified in ASME B 56.1. Nonconformance to 3.12.2 or 3.12.2.1 and ASME B 56.1, except a 20 percent slope is required, shall constitute failure of this test.

4.5.2.2.9 Lifting speed and lowering speed. Demonstrate each truck in accordance with method no. 2. Nonconformance to 3.24.1 or 3.24.2 shall constitute failure of this test.

4.5.2.2.10 Right angle turn. Demonstrate the truck in accordance with test method no. 3. Dimension “A” greater than specified in the applicable specification sheet shall constitute failure of this test (see 3.24.3).

4.5.2.2.11 Travel speed. Demonstrate each truck in accordance with test method no. 4. Nonconformance to 3.24.4 shall constitute failure of this test.

4.5.2.2.12 Slope ascension. Demonstrate each truck in accordance with test method no. 5. Nonconformance to 3.24.5 shall constitute failure of this test.

4.5.2.2.13 Tilt. Demonstrate the truck in accordance with test method no. 6. Nonconformance to 3.24.6 shall constitute failure of this test.

4.5.2.2.14 Upright height, maximum lift height, free lift height. Demonstrate the truck in accordance with test method no. 1. Nonconformance to 3.24.7, 3.24.8, or 3.24.9 shall constitute failure of this test.

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4.5.2.2.15 Underclearance. The forklift shall be tested in accordance with 3.24.10. Nonconformance to 3.24.10 shall constitute failure of this test.

4.5.2.2.16 Drift. Demonstrate the truck in accordance with test method no. 7. Nonconformance to 3.24.11 shall constitute failure of this test.

4.5.2.2.17 Stability. Conduct "forward stacking", "forward travel", "lateral stacking" and "lateral travel" tilting platform tests in accordance with ASME B 56.1. Nonconformance to 3.4.12 shall constitute failure of this test.

4.5.2.2.18 Truck operations. Demonstrate the truck in accordance with test method no. 8. Nonconformance to 3.24.13 shall constitute failure of this test.

4.5.2.2.19 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474, requirements and reported in the format indicated by MIL-STD-1474 figure 7. As a minimum: noise levels shall be measured when equipment is operating under full load. The noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with MIL-STD-1474 provisions shall constitute failure of this test.

4.5.2.2.20 Tire loading. Determine the weight supported by the steer (trail) wheels with and without a rated load on the forks by driving the wheels onto a platform scale or utilizing two axle scales. Determine the weight supported by drive (load) wheels in the same manner as the steer wheels. Nonconformance to 3.28 shall be cause for rejection.

4.5.2.2.21 Electromagnetic radiation. Test the truck *in* accordance with SAE J551. As a minimum, the electromagnetic radiation report shall include results of all testing and a list of components used to reduce the electromagnetic radiation. Nonconformance to 3.29 shall constitute failure of this test.

4.5.2.2.22 Maintenance operations. Perform the maintenance operations as listed in 3.33.1. Nonconformance shall constitute failure of this test.

4.5.2.2.23 Inspection of transportability.

4.5.2.2.23.1 First article inspection of transportability.

4.5.2.2.23.1.1 Examination. The first article shall be examined for the defects listed in 4.5.2.2.23.2.1. The presence of one or more defects shall be cause for rejection. Any defect(s) shall be corrected and the first article re-examined.

4.5.2.2.23.1.2 Tests. The first article shall be tested as follows:

4.5.2.2.23.1.2.1 Tiedown provisions test. The tiedown provisions shall be tested in accordance with MIL-STD-209 to prove conformance to 3.32.1. Inability to meet the requirements of 3.32.1 shall constitute failure of this test.

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4.5.2.2.23.1.2.2 Slinging provisions test. The slinging provisions shall be tested in accordance with MIL-STD-209 to prove conformance to 3.32.2. Inability to meet the requirements of 3.32.2 shall constitute failure of this test.

4.5.2.2.23.1.2.3 Rail impact test. The first article truck shall be tested in accordance with MIL-STD-810 to prove conformance to 3.32.3. Inability to meet the requirements of 3.32.3 shall constitute failure of this test.

4.5.2.2.23.1.2.4 Highway transport test. Inability to provide data demonstrating that the truck meets the requirements of 3.32.4 shall constitute failure of this test.

4.5.2.2.23.1.2.5 Water transportability test. Inability to meet the requirements of 3.32.5 shall constitute failure of the test.

4.5.2.2.23.2 Quality conformance inspection of transportability.

4.5.2.2.23.2.1 Examination. Samples selected in accordance with 4.4.1 shall be examined for the following defects. The presence of one or more defects shall be cause for rejection.

157. Tiedown provisions not as specified (3.32.1).
158. Slinging provisions not as specified (3.32.2).

4.5.2.2.24 Endurance/reliability testing. The truck carrying the capacity load shall be operated for a minimum of 320 hours over a course as shown in accordance with method No. 8. Hours specified here shall be productive clock hours excluding fueling, servicing, and downtime. During this test, the truck shall operate for a minimum of 8 hours without refueling as specified in 3.7.1.4.

Accept/Reject Criteria.

- Accept with zero (0) endurance failure, reject otherwise.
- Accept with five (5) or less reliability failures. Reject with six (6) or more reliability failures.
- Relevant failures shall be determined by authorized government representative. Nonconformance shall constitute failure of this test.

4.5.2.2.24.1 Endurance failure. Any incident resulting in internal component failure of the engine transmission, transfer case, differential hydraulic pump or permanent deformation of the truck chassis shall constitute an endurance failure.

4.5.2.2.24.2 Reliability failure. A reliability failure is defined as any malfunction which causes degradation of specified operations or any other unscheduled maintenance action to be taken in order to restore the truck to specified requirements. Those malfunctions resulting from accidents, noncompliance with prescribed operational or maintenance procedures, faulty test instrumentation, fail safe devices, operating as specified shall be considered non relevant failures. Non repetitive replacement of light bulbs, fuses, tires and tightening of fasteners shall be considered non relevant failures. Any impending malfunction detected during test which constitutes a

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safety hazard to secondary damage either to the item under test or testing instrumentation shall be considered as a relevent failure.

4.5.2.2.25 Operation with MIL-T-83133. When testing for operation with MIL-T-83133 is required (see 6.2), the following test shall be conducted. Following completion of all previously specified tests for the first article truck(s), drain and flush the entire fuel system, refill with fuel conforming to MIL-T-83133 and subject the truck(s) to the travel speed (see 4.5.2.2.11), slope ascension (see 4.5.2.2.3.12), and endurance/reliability test (4.5.2.2.24). Failure of any tests or nonconformance to 3.7 will be cause for rejection.

4.6 Inspection of packing. The preservation, packing and marking for the forklift truck shall be examined and tested to determine compliance with the applicable requirements of MIL-STD-162.

5. PACKAGING

5.1 Preservation, packing, and marking. The level of preservation shall be level A, B, C, or Commercial as specified (see 6.2). The level of packing shall be level A, B or Commercial as specified (see 6.2). The forklift truck shall be preserved, packed, and marked in accordance with the applicable requirements of MIL-STD-162, type 1 (mobile). For all levels, the engine cooling system shall be preserved using the water/anti-freeze procedure of MIL-E-10062.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The truck described herein are intended for stacking, unstacking, and moving cargo in and around warehouses, loading platforms and docks within the military supply system. The truck is intended for operation over paved, semiprepared and other hard surfaces for short distances.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number, and date of the specification.
b. Specification sheet title, number, and date.
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. Time frame required for submission of the first article truck(s) and number of trucks required (see 3.2).
e. When the Government will conduct any or all of the first-article truck examination and tests. When the Government will conduct some but not all of the first article truck examination and tests, the contracting officer shall specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
f. When ambient temperature is required to be -25 to 120 °F (see 3.4).
g. When a type DS truck is required (see 3.5, 3.26.3).

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h. When nuclear safety is required (see 3.5.1, 3.5.2, 3.9, 3.10, 3.12.2.1, 3.24.2, and 3.24.6).
i. When a rearview mirror is required (see 3.5.3).
j. When a transmission oil sampling valve shall be furnished (see 3.7.1.8).
k. When a personnel cab is required (see 3.22).
l. When the portions of the truck normally painted, both internally and externally, shall be cleaned, treated, and painted in accordance with MIL-T-704, type F for ferrous metals and type G for nonferrous metals fiberglass top coat camouflage green 383 (see 3.25).
m. When other than operational use is required, specify service and adjustment required (see 4.5.2.2.5.4).
n. Level of preservation, packaging, and packing required (see 5.1).
o. When another color paint is required (see 3.25).
p. When electromagnetic testing is required (see 3.29).
q. When technical information package (TIP) is required (see 3.30).
r. When truck operations of only 24 hours are required (see 3.24.13).
s. When warning light is required (see 3.20.4).
t. When slinging and tiedown provisions are required (see 3.32).
u. When M16 mounting is not required (see 3.35)
v. When servicing and adjustment are not required prior to delivery (see 4.5.2.2.5.4).
w. When first article test shall include a test for operation with MIL-T-83133 (see 3.7).
x. When truck operations test is not required (see 3.24.13).
y. When samples are not subjected to examination (see 4.4.2).
z. When each truck in lot shall be examined (see 4.4.2).
aa. When samples are not subjected to sample tests (see 4.4.3.2).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Fail-safe hydraulic lift system. A system with inherent safety features which will automatically prevent load lowering in the event of hydraulic line failure and provide a means of manually overriding the system to safely and slowly lower the load or empty carriage so that repairs can be made without incident.

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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6.6 Subject term (key word) listing.
Diesel
Fork
Lift
Trucks

Custodians:
Army - ME
Navy - SA
Air Force – 99

Review activity:
Army - SM
Navy - OS
DLA - CS

User activities:
Army – MT
Navy - MC

Preparing activity
Army - ME

Project 3930-0651

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APPENDIX

FOR

TRUCKS, LIFT, FORK, DIESEL

10. SCOPE. The test methods contained in this appendix are to determine whether trucks procured under this specification conform to the requirements set forth.

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TEST METHOD NO. 1

MAXIMUM FORK HEIGHT - FREE LIFT HEIGHT

COLLAPSED MAST HEIGHT

1. Test Course:

   Level flat surface:

2. Test Procedure:

   a. Measure the true vertical, distance from floor to uppermost projection of the upright assembly with mast in true vertical position. (This is the actual collapsed mast height.)
   b. Raise rated load in a horizontal position until the inner mast begins to exceed the specified collapsed mast height.
   c. Measure and record the true vertical distance to the top surface of the forks at the specified load center. (This is free lift height.)
   d. Adjust rated load to 12 inches, ±4 inches, above ground level and position load in true horizontal position.
   e. Attach a plumb bob to the rated load at the load center at the bottom and mark a reference point on the ground.
   f. Elevate the test load to the maximum height and check the reference point. If necessary, reposition the mast so that the plumb bob is directly over the original reference point.
   g. Measure and record the true vertical distance from floor to the top surface of the forks. (This is maximum fork height.)

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TEST METHOD NO. 2

LIFTING SPEED - LOWERING SPEED

1. Test Procedure:

   a. Measure and record distance in inches from floor to top surface of forks in fully lowered position.
   b. Record time in seconds required to raise rated load to maximum fork height using timing device. Record the difference between maximum fork height and the height of the forks in the lowered position.
   c. Record time to lower rated load at maximum speed (full open valve) to 3- to 4-foot height. Load to be abruptly stopped at the 3- to 4-foot height.
   d. Repeat steps (b) and (c) a minimum of three times, except no measurements are required.
   e. When nuclear safety is required, determine lowering speed immediately before forks contact ground. If a single stage lift cylinder is provided, no additional demonstrating is required but lowering speed determined above shall not exceed 65 feet per minute. If a multistage lift cylinder is provided, forks with rated load shall be lowered until further lowering will cause constant speed descent. Record time to fully lower from this point and the distance traveled by forks. Calculated lowering speed shall not exceed 65 feet per minute.

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TEST METHOD NO. 3

RIGHT ANGLE TURN

1. Test Course:

   a. Level paved surface.
   b. Test course in accordance with "course layout" (see figure A-1).

2. Test Procedure:

   a. Truck is placed with the most forward edge of centered rated load coincident to line "B" (see course layout). When forks are longer than rated load, the fork tips shall touch line “B” and the fork length is included in the “A” dimension. Mast in vertical position.
   b. Pallet is raised 6 inches above floor.
   c. Position movable vertical barrier within 4 inches of counterweight of truck.
   d. Turn steer wheels to the extreme left position and back truck through 90-degree turn.
   e. Measure the perpendicular distance from line "B" to the position assumed by the movable barrier. This is dimension "A".
   f. Return truck to initial position.
   g. Turn steer wheels to extreme right and repeat (a) through (e).

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TEST METHOD NO. 4

SPEED

1. Test Course:

   Test course shall be a dry, level surface of sufficient length to attain specified speed prior to entering measured distance, plus measured distance, plus sufficient stopping distance.

2. Test Procedure:

   a. Operate the truck at governed speed with rated load in carry position.
   b. Drive the truck a sufficient distance to attain specified speed prior to entering the measured test course.
   c. Record length of measured distance, time to traverse measured distance. Measured distance shall be equal to or greater then 22 feet.
   d. Repeat for a total of six runs, three in each direction for each range, except only time measurement is required. Repeat total of six runs in reverse gear, three in each direction for each range, except only time measurement is required. Travel speed in each gear shall be the average of the six runs.

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Test Method No. 3

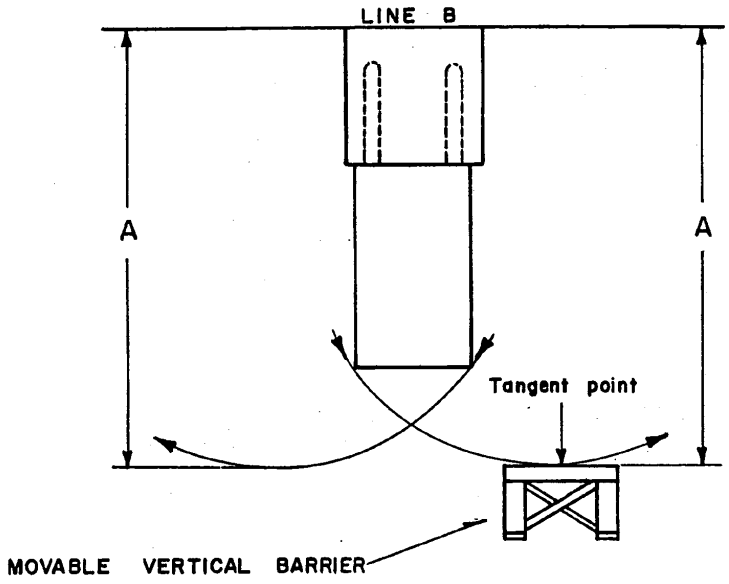


Figure A-1. COURSE LAYOUT

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TEST METHOD NO. 5

SLOPE ASCENSION

1. Test Course:

   Tilt table or ramp.

2. Test Procedures:

   a. Drive truck forward with rated load along horizontal approach and onto ramp of specified percent grade and bring to complete stop. (All wheels supported on the slope.)
   b. Release brakes and start from dead stop and proceed up ramp.

3. Optional Procedure:

   a. Determine gross weight of truck with rated load (GVW).
   b. Attach a tension dynamometer to the rear towing eye and to suitable ground anchor.
   c. Put transmission in forward low range.
   d. Accelerate engine until a minimum force equal to 30 percent GVW is obtained at the towing eye.

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TEST METHOD NO. 6

UPRIGHT TILT

1. Test Course:

Level, flat surface.

2. Test Procedures:

a. Tilt.

(1) Place truck without load on flat surface.
(2) Raise forks to an elevation of 2 feet above the surface, tilt boom as far forward as possible, and record angle of tilt.
(3) Tilt boom as far rearward as possible and record angle of tilt.

b. Tilt cavitation.

(1) Tie down rear of the truck to prevent overturning.
(2) Install pressure gage to read hydraulic pressure at head end of tilt cylinder.
(3) Raise rated load to maximum lift height with mast vertical.
(4) Without adjustment of height control, tilt load to full forward tilt at engine low idle by opening forward tilt valve completely (no feathering of valve is permitted).
(5) Record lowest pressure reading at tilt cylinder.
(6) Repeat b(3), (4), and (5) by feathering valve to produce slowest tilt speed practical.
(7) Record lowest pressure reading at tilt cylinder.

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TEST METHOD NO. 7

DRIFT

1. Test Course:

   Level, flat surface.

2. Test Procedure:

   a. Heat the hydraulic oil by exercising the lift and tilt functions.
   b. Raise rated load to maximum fork height obtainable with outer channel in true vertical position. Mark reference point on mast upper stage.
   c. Record initial outer mast tilt.
   d. Hold for specified time and record downward drift and rotational drift.

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TEST METHOD NO. 8

TRUCK OPERATIONS

1. Test course:

   a. The test course shall be a level, improved surface laid out in accordance with figure A-2 or equivalent length course if a 200 ft. diameter pad is not available. Station 1 and station 3 shall be platforms that are not less than 8 feet in height and capable of supporting rated load. Station 2 shall be at ground level.
   b. Obstacles shall be in accordance with figure A-3 and placed so that the nearest obstacle is not less than 14 feet from station 1.

2. Test Procedure:

   a. Initially, place rated loads on platforms at stations 1 and 3.
   b. Pick up load from station 1; lower the rated load 12 inches maximum above ground; position the mast to maximum rearward tilt; back up truck being sure all wheels travel over obstacles; travel forward and deposit load at station 2.
   c. Disengage load at station 2; back out; then travel forward to station 3. Prior to engaging load at station 3, raise mast to maximum lift height to activate relief valve. Lower forks and engage load.
   d. Pick up load at station 3; lower the rated load 12 inches maximum above ground; position the mast to maximum rearward tilt; back out; then travel forward and deposit load at station 1, being sure all wheels travel over obstacles.
   e. Disengage load at station 1; back out, being sure all wheels travel over obstacles; then travel forward to station 2.
   f. Pick up load at station 2; raise the rated load to 12 inches above ground; position the mast to maximum rearward tilt; back out; then travel forward and deposit load at station 3.
   g. Steps 2(b) through 2(f) shall constitute one circuit. For 50 percent of the circuits, station 2 and Station 3 shall be interchanged so that they are in clockwise position rather then in their initial counterclockwise position. Stations shall be changed at the end of each 8-hour operating day.
   h. Truck with rated load shall be operated for the required number of hours.
   i. Record total circuits for each hour of operation.
   j. Truck shall complete 6 to 8 circuits per hour.

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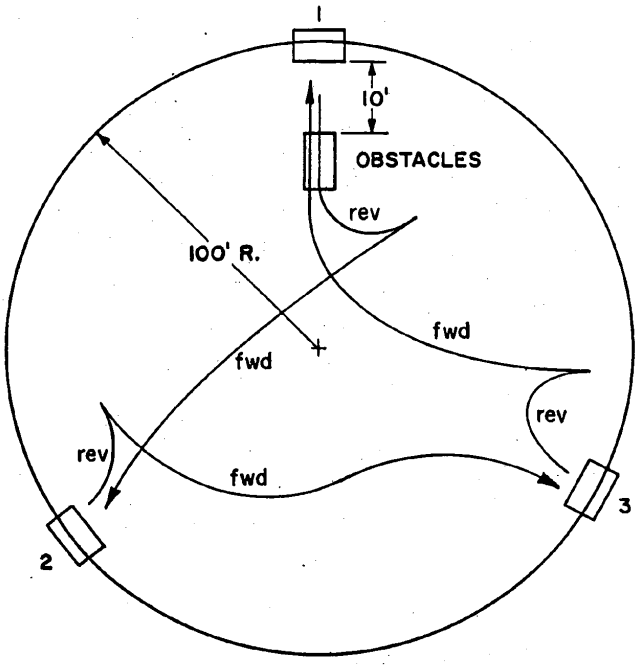
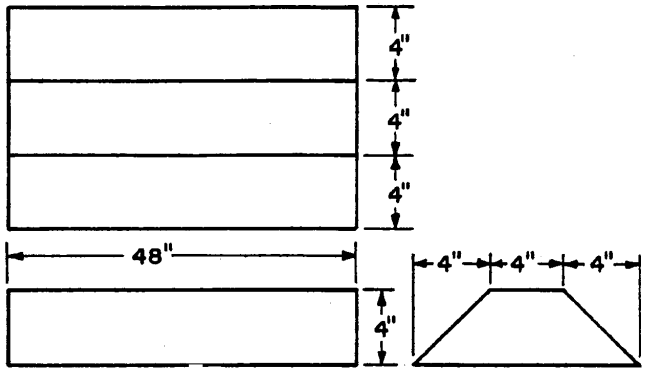


Figure A-2 TRUCK OPERATIONS COURSE

X-2347B

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LAYOUT OF OBSTACLE COURSE

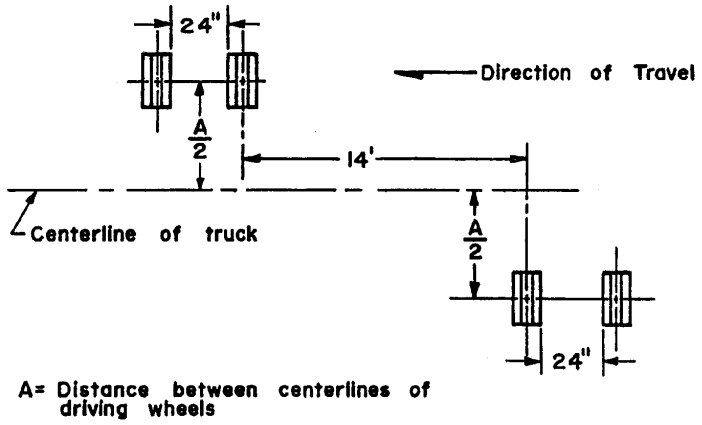


Figure A-3 DETAILS OF OBSTACLE CONSTRUCTION

X-2348B

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter must be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-T-52864D	2. DOCUMENT DATE (YYMMDD) 900906

3. DOCUMENT TITLE
TRUCKS, LIFT, FORK: DUSEL-ENGINE-DRIVEN, PNEUMATIC-TIRED 10,000, 15,000 AND 20,000 POUND CAPACITY; GENERAL SPECIFICATION FOR

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) DSN (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME USA TACOM, AMSTA-TR-E-ELE/21 (4EELD)	b. TELEPHONE (Including Area Code) (1) Commercial (810)574-6156 (2) DSN 786-6156
c. ADDRESS (Include Zip Code) US Army Tank-automotive and Armaments Command Attn: AMSTA-TR-E-ELE/21 Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 681-9340 DSN 761-9340

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MIL-T-52864/1A
1 June 1983
SUPERSEDING
MIL-T-52864/1
16 June 1975

MILITARY SPECIFICATION SHEET

TRUCK, LIFT, FORK: DIESEL-ENGINE-DRIVEN, PNEUMATIC-TIRED,

10,000 POUND CAPACITY AT 24 INCH

LOAD CENTER, 168 INCH LIFT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the truck described herein shall consist of this document and the latest issue of MIL-T-52864.

REQUIREMENTS

1. Capacity: 10,000 pounds at 24-inch load center.
2. Collapsed mast height: 130 inches maximum.
3. Maximum fork height: 168 inches minimum.
4. Free lift height: As applicable.
5. Fork size: 2-9/16 inches maximum thickness X 7-1/16 inches maximum width X 48 inches long, plus 0 inch minus 1/2 inch.
6. Fork spacing: 18 inches to minimum 56 inches.
7. Overhead guard height: 105 inches maximum.
8. Right-angle turn dimension "A": 210 inches maximum.
9. Underclearane beneath mast assembly: 5 inches minimum.
10. Transmission: Not less than two speeds forward and two speeds reverse.

Custodian:
Army – ME
Navy - SA
Air Force – 99

Preparing activity:
Army - ME

Project 3930-0500

Review activity:
Army - SM

User activities:
Army - MT
Navy - MC, SH

FSC 3930

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MIL-T-52864/2B
22 April 1987
SUPERSEDING
MIL-T-52864/2A
1 June 1983

MILITARY SPECIFICATION SHEET

TRUCK, LIFT, FORK: DIESEL-ENGINE-DRIVEN, PNEUMATIC-TIRED,

15,000 POUND CAPACITY AT 24-INCH

LOAD CENTER, 100-INCH LIFT

The complete requirements for acquiring the truck described herein shall consist of this specification and the latest issue of MIL-T-52864.

REQUIREMENTS

1. Capacity: 15,000 pounds at 24-inch load center.
2. Collapsed mast height: 95 inches maximum.
3. Maximum fork height: 100 inches minimum.
4. Free lift height: As applicable.
5. Fork size: 2-9/16 inches maximum thickness X 8-1/16 inches maximum width X 48 inches long, +0 inch -1/2 inch.
6. Fork spacing: 18 inches to minimum 60 inches.
7. Overhead guard height: 116 inches maximum.
8. Right-angle turn dimension "A": 240 inches maximum.
9. Underclearance beneath mast assembly: 6 inches minimum.
10. Transmission: Not less than two speeds forward and two speeds reverse.

Custodian:
Army – ME
Navy - SA
Air Force – 99

Preparing activity:
Army - ME

Project 3930-0625

Review activity:
Army – SM
Navy - OS

User activities:
Army - MT
Navy - MC

AMSC N/A FSC 3930

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited

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INCH-POUND

MIL-T-52864/3C
23 March 1990
SUPERSEDING
MIL-T-52864/3B
22 April 1987

MILITARY SPECIFICATION SHEET

TRUCK, LIFT, FORK: DIESEL-ENGINE-DRIVEN, PNEUMATIC-TIRED, 15,000-POUND CAPACITY AT 24-INCH LOAD CENTER, 210-INCH LIFT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the truck described herein shall consist of this specification and the latest issue of MIL-T-52864.

REQUIREMENTS

1. Capacity: 15,000 pounds at 24-inch load center.
2. Collapsed mast height: 154 inches maximum.
3. Maximum fork height: 210 inches minimum.
4. Free lift height: As applicable.
5. Fork size: 2-9/16 inches maximum thickness X 8-1/16 inches maximum width X 48 inches long, +0 inch -1/2 inch.
6. Fork spacing: 18 inches to minimum 60 inches.
7. Overhead guard height: 116 inches maximum.
8. Right-angle turn dimension "A": 240 inches maximum.
9. Underclearane beneath mast assembly: 6 inches minimum.
10. Transmission: Not less than two speeds forward and two speeds reverse.

Custodian:
Army – ME
Navy - SA
Air Force – 99

Preparing activity:
Army - ME

Project 3930-0648

Review activity:
Army – SM
Navy – OS
Air Force - 84

User activities:
Army - MT
Navy - SH

AMSC N/A

FSC 3930

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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MIL-T-52864/4A
1 June 1983
SUPERSEDING
MIL-T-52864/4
16 June 1975

MILITARY SPECIFICATION SHEET

TRUCK, LIFT, FORK: DIESEL-ENGINE-DRIVEN, PNEUMATIC-TIRED, 20,000-POUND CAPACITY AT 24-INCH LOAD CENTER, 120-INCH LIFT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the truck described herein shall consist of this document and the latest issue of MIL-T-52864.

REQUIREMENTS

1. Capacity: 20,000 pounds at 24-inch load center.
2. Collapsed mast height: 117 inches maximum.
3. Maximum fork height: 120 inches minimum.
4. Free lift height: As applicable.
5. Fork size: 3-1/16 inches maximum thickness X 10-1/16 inches maximum width X 72 inches long, plus 0 inch minus 1/2 inch.
6. Fork spacing: 22 inches to minimum 64 inches.
7. Overhead guard height: 116 inches maximum.
8. Right-angle turn dimension "A": 272 inches maximum.
9. Underclearane beneath mast assembly: 8 inches minimum.
10. Transmission: Not less than three speeds forward and three speeds reverse.

Custodian:
Army – ME
Navy - SA
Air Force – 99

Preparing activity:
Army - ME

Project 3930-0503

Review activity:
Army - SM

User activities:
Army - MT
Navy - MC, SH

FSC 3930

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MIL-T-52864/5A
1 June 1983
SUPERSEDING
MIL-T-52864/5
16 June 1975

MILITARY SPECIFICATION SHEET

TRUCK, LIFT, FORK: DIESEL-ENGINE-DRIVEN, PNEUMATIC-TIRED,

20,000 POUND CAPACITY AT 24-INCH

LOAD CENTER, 210-INCH LIFT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the truck described herein shall consist of this document and the latest issue of MIL-T-52864.

REQUIREMENTS

1. Capacity: 20,000 pounds at 24-inch load center.
2. Collapsed mast height: 165 inches maximum.
3. Maximum fork height: 210 inches minimum.
4. Free lift height: As applicable.
5. Fork size: 3-1/16 inches maximum thickness X 10-1/16 inches maximum width X 72 inches long, plus 0 inch minus 1/2 inch.
6. Fork spacing: 22 inches to minimum 64 inches.
7. Overhead guard height: 116 inches maximum.
8. Right-angle turn dimension "A": 272 inches maximum.
9. Underclearane beneath mast assembly: 8 inches minimum.
10. Transmission: Not less than three speeds forward and three speeds reverse.

Custodian:
Army – ME
Navy - SA
Air Force – 99

Preparing activity:
Army - ME

Project 3930-0504

Review activity:
Army - SM

User activities:
Army - MT
Navy - MC, SH

FSC 3930

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MIL-T-52864/6A
1 June 1983
SUPERSEDING
MIL-T-52864/6
16 June 1975

MILITARY SPECIFICATION SHEET

TRUCK, LIFT, FORK, DIESEL-ENGINE-DRIVEN, PNEUMATIC-TIRED,

10,000-POUND CAPACITY AT 48-INCH

LOAD CENTER, 150-INCH LIFT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirement for procuring the truck described herein shall consist of this document and the latest issue of MIL-T-52864.

REQUIREMENTS

1. Capacity: 10,000 pounds at 48-inch load center.
2. Collapsed mast height: 95 inches maximum.
3. Maximum fork height: 150 inches minimum.
4. Free lift height: 25 inches minimum.
5. Fork size: 2-9/16 inches maximum thickness X 8-1/16 inches maximum width X 72 inches long, plus 0 inch minus 1/2 inch.
6. Fork spacing: 18 inches to minimum 60 inches.
7. Overhead guard height: 95 inches maximum.
8. Right-angle turn dimension "A": 290 inches maximum.
9. Underclearance beneath mast assembly: 6 inches minimum.
10. Transmission: Not less than two speeds forward and two speeds reverse.

Custodians:
Army – ME
Navy - SA
Air Force – 99

Preparing activity
Army - ME

Project 3930-0505

Review activity:
Army - SM

User activities
Army - MT
Navy - MC, SH

FSC 3930

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MIL-T-52864/7A
1 June 1983
SUPERSEDING
MIL-T-52864/7
16 June 1975

MILITARY SPECIFICATION SHEET

TRUCK, LIFT, FORK: DIESEL-ENGINE-DRIVEN, PNEUMATIC-TIRED,

15,000-POUND CAPACITY AT 48-INCH

LOAD CENTER, 210-INCH LIFT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the truck described herein shall consist of this document and the latest issue of MIL-T-52864.

REQUIREMENTS

1. Capacity: 15,000 pounds at 48-inch load center.
2. Collapsed mast height: 165 inches maximum.
3. Maximum fork height: 210 inches minimum.
4. Free lift height: As applicable.
5. Fork size: 3-1/16 inches maximum thickness X 10-1/16 inches maximum width X 72 inches long, plus 0 inch minus 1/2 inch.
6. Fork spacing: 32 inches to minimum 80 inches.
7. Overhead guard height: 116 inches maximum.
8. Right-angle turn dimension “A”: 312 inches maximum.
9. Underclearance beneath mast assembly: 10 inches minimum.
10. Transmission: Not less than three speeds forward and three speeds reverse.

Custodians:
Army – ME
Navy - SA
Air Force – 99

Preparing activity:
Army - ME

Project 3930-0506

Review activity:
Army - SM

User activities:
Army - MT
Navy - MC, SH

FSC 3930

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